Bayesian Neural Network Priors Revisited

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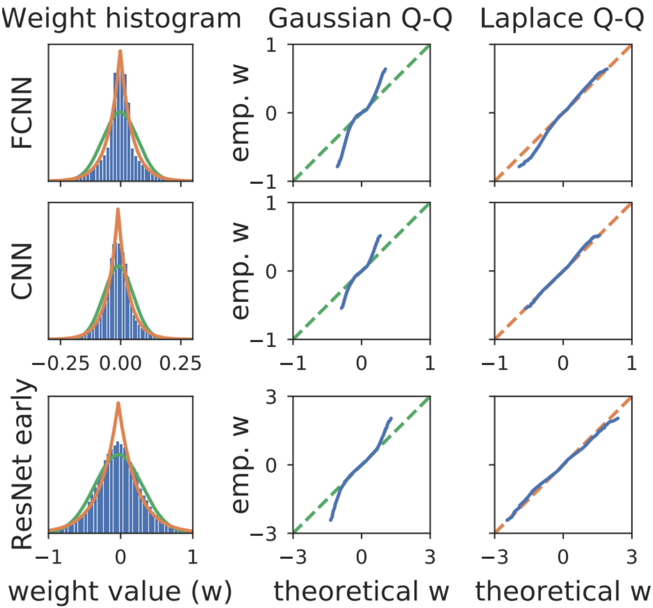
Main Idea

- **Problem**: Isotropic Gaussian priors are the de facto standard for Bayesian Neural Networks (BNNs), but may not reflect true beliefs about weight distributions
- **Observation**: Networks trained with SGD show systematic patterns in weight distributions that differ from isotropic Gaussians
- **Hypothesis**: Building these empirical patterns into priors can improve BNN performance
- Better priors may help explain/mitigate the "cold posterior effect" (improved performance with tempered posteriors)

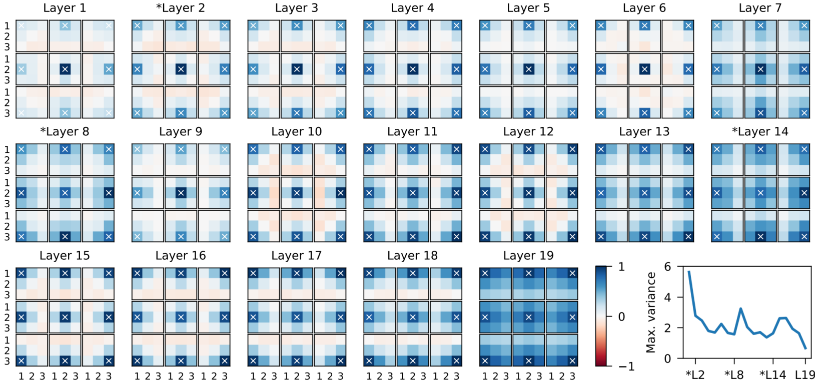
Current Method: Gaussian Priors

- **Isotropic Gaussian priors** are used across almost all Bayesian deep learning.
- **Advantages**:
  - Simple to implement and optimize
  - Maximum entropy for given variance
  - Well-studied theoretical properties
- **Problems**:
  - May not reflect true weight distributions
  - Can lead to poor predictive performance vs SGD
  - May contribute to cold posterior effect

Empirical Weight Distributions in FCNNs



Spatial Correlations in CNN Weights



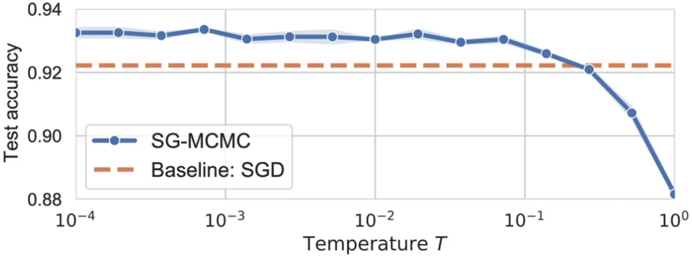
- CNN/ResNet weights show **strong spatial correlations**
- Nearby pixels in filters are highly correlated
- Correlation strength increases in deeper layers

The Cold Posterior Effect

- **Tempered posterior**:

$$\log p(w|x,y)^{\frac{1}{T}} = \frac{1}{T}[\log p(y|w,x) + \log p(w)] + Z(T)$$

  - $T = 1$: Standard Bayesian posterior
  - $T < 1$: "Cold posterior" (overcounting data)
  - $T > 1$: "Warm posterior" (undercounting data)
- If prior and likelihood are well-specified, $T = 1$ should be optimal

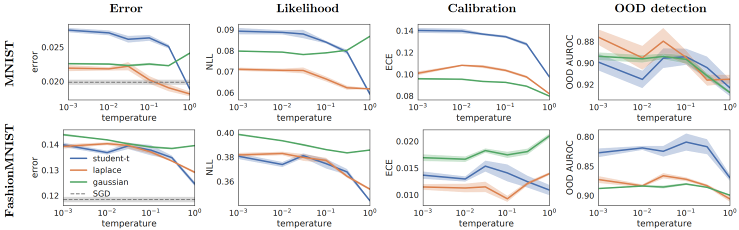


Alternative Prior Distributions

- **Heavy-tailed priors for FCNNs**:
  - Laplace distribution: $p(x) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$
  - Student-t distribution: adjustable tail heaviness via ν parameter
- **Correlated Gaussian priors for CNNs/ResNets**:
  - Multivariate Gaussian with block-diagonal covariance
  - Matern kernel ($\nu = 1/2$) for spatial correlations:

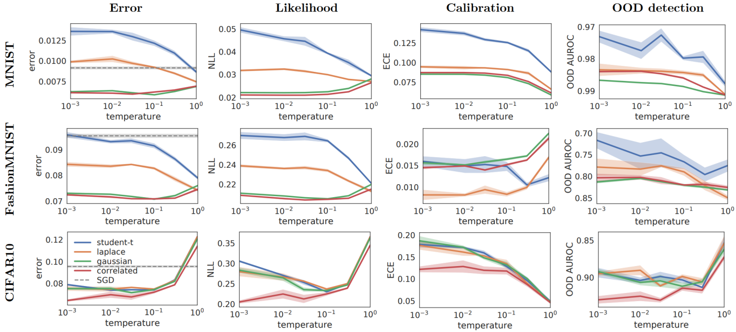
$$\mathsf{cov}(w_{i,j}, w_{i',j'}) = \sigma^2 \exp\left(\frac{-d(j,j')}{\lambda}\right)$$

Performance: FCNNs with Different Priors



- Heavy-tailed priors outperform Gaussian in test error/NLL
- Eliminate cold posterior effect in FCNNs

Performance: CNNs/ResNets with Different Priors



- Correlated Gaussian priors outperform isotropic ones
- Heavy-tailed priors reduce cold posterior effect but hurt performance
- Correlated priors *increase* cold posterior effect in ResNets

Key Takeaways

- **Empirical analysis matters**: SGD-trained networks show systematic patterns that should inform prior choice
- **Architecture-dependent priors**:
  - FCNNs: Heavy-tailed priors (Laplace/Student-t)
  - CNNs/ResNets: Spatially correlated Gaussian priors
- **Cold posterior effect**:
  - In FCNNs: Likely due to prior misspecification (fixed by heavy-tailed priors)
  - In ResNets: More complex (possibly likelihood/data augmentation issues)